

Implementation of Network Scan Detection with Suricata and Wazuh

Objective

The objective of this lab was to detect Nmap reconnaissance activity against a Debian 13 system using Suricata and forward the generated alerts to Wazuh for centralized monitoring and alerting, also Test File Integrity Management in Wazuh.

Lab Architecture

The environment consisted of:

- Attacker Machine (Nmap)
 - Debian 13 Target System
 - Suricata IDS
 - Wazuh Agent
 - Wazuh Manager and Dashboard
-

Environment Details

Component	Version
Operating System	Debian GNU/Linux 13 (Trixie)
IDS	Suricata
SIEM	Wazuh
Scanner	Nmap
Virtualization	VirtualBox
Network Configuration	NAT + Host-Only Adapter

Step 1 – Verify Network Interfaces

The target machine was configured with:

- Adapter 1: NAT
- Adapter 2: Host-Only Network

Network interfaces were verified using:

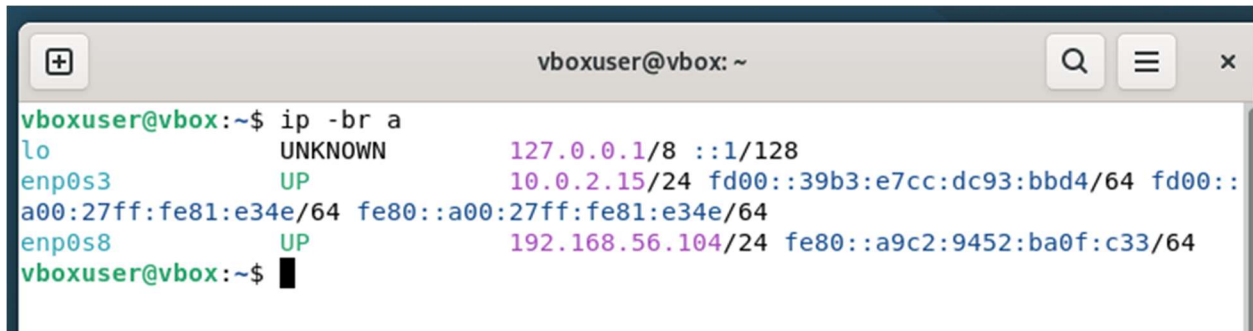
`ip -br a`

Example:

```
enp0s3  UP 10.0.2.15/24 (NAT)
enp0s8  UP 192.168.56.102/24 (Host Only)
```

The Host-Only interface was used for scan detection testing.

Screenshot Placeholder

A screenshot of a terminal window titled 'vboxuser@vbox: ~'. The terminal shows the command 'ip -br a' being executed. The output is as follows:

```
vboxuser@vbox:~$ ip -br a
lo                UNKNOWN      127.0.0.1/8 ::1/128
enp0s3            UP          10.0.2.15/24 fd00::39b3:e7cc:dc93:bbd4/64 fd00::
a00:27ff:fe81:e34e/64 fe80::a00:27ff:fe81:e34e/64
enp0s8            UP          192.168.56.104/24 fe80::a9c2:9452:ba0f:c33/64
vboxuser@vbox:~$
```

Step 2 – Install Suricata

Update package repositories:

```
sudo apt update
```

Install Suricata:

```
sudo apt install -y suricata suricata-update
```

Verify installation:

```
suricata --build-info
systemctl status suricata
```

Screenshot Placeholder

```
root@vbox:/home/vboxuser# suricata --build-info
This is Suricata version 7.0.10 RELEASE
Features: NFO PCAP SET_BUFF AF_PACKET HAVE_PACKET_FANOUT LIBCAP_NG LIBNET1.1 HAVE_HTTP_URI_NORMALIZE_HOOK PCRE_JIT HAVE_NSS HTTP2_DECOMPRESSION HAVE_LUA HAVE_JA3 HAVE_JA4 HAVE_LUAJIT
HAVE_LIBJANSSON TLS TLS_C11 MAGIC RUST POPCNT64
SIMD support: SSE_4_2 SSE_4_1 SSE_3 SSE_2
Atomic intrinsics: 1 2 4 8 16 byte(s)
64-bits, Little-endian architecture
GCC version 14.2.0, C version 201112
compiled with _FORTIFY_SOURCE=2
L1 cache line size (CLS)=64
thread local storage method: _Thread_local
compiled with LibHTP v0.5.50, linked against LibHTP v0.5.50

root@vbox:/home/vboxuser# systemctl status suricata
● suricata.service - Suricata IDS/IDP daemon
   Loaded: loaded (/usr/lib/systemd/system/suricata.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-08-08 05:26:41 UTC; 15min ago
     Invocation: 7909334bbc544fd0b37f3941430f4730
       Docs: man:suricata(8)
             man:suricatasc(8)
             https://suricata.io/documentation/
    Main PID: 831 (Suricata-Main)
      Tasks: 8 (limit: 2281)
     Memory: 530.2M (peak: 547.7M)
        CPU: 1min 7.321s
    CGroup: /system.slice/suricata.service
            └─831 /usr/bin/suricata -D --af-packet -c /etc/suricata/suricata.yaml --pidfile /run/suricata.pid

Aug 08 05:26:40 vbox systemd[1]: Starting suricata.service - Suricata IDS/IDP daemon...
Aug 08 05:26:41 vbox suricata[774]: i: suricata: This is Suricata version 7.0.10 RELEASE running in SYSTEM mode
Aug 08 05:26:41 vbox systemd[1]: Started suricata.service - Suricata IDS/IDP daemon.
```

Step 3 – Download and Update Detection Rules

Suricata rules were updated using:

```
sudo suricata-update
```

Restart Suricata:

```
sudo systemctl restart suricata
```

Purpose:

- Download Emerging Threats rules
- Enable detection of reconnaissance activity
- Keep signatures updated

Screenshot Placeholder

```
root@vbox: /home/vboxuser# suricata-update
8/8/2026 -- 05:42:56 - <Info> -- Using data-directory /var/lib/suricata.
8/8/2026 -- 05:42:56 - <Info> -- Using Suricata configuration /etc/suricata/suricata.yaml
8/8/2026 -- 05:42:56 - <Info> -- Using /etc/suricata/rules for Suricata provided rules.
8/8/2026 -- 05:42:56 - <Info> -- Found Suricata version 7.0.10 at /usr/bin/suricata.
8/8/2026 -- 05:42:56 - <Info> -- Loading /etc/suricata/suricata.yaml
8/8/2026 -- 05:42:57 - <Info> -- Disabling rules for protocol pgsq
8/8/2026 -- 05:42:57 - <Info> -- Disabling rules for protocol modbus
8/8/2026 -- 05:42:57 - <Info> -- Disabling rules for protocol dnp3
8/8/2026 -- 05:42:57 - <Info> -- Disabling rules for protocol enip
8/8/2026 -- 05:42:57 - <Info> -- No sources configured, will use Emerging Threats Open
8/8/2026 -- 05:42:57 - <Info> -- Checking https://rules.emergingthreats.net/open/suricata-7.0.10/emerging.rules.tar.gz.md5.
8/8/2026 -- 05:42:58 - <Info> -- Fetching https://rules.emergingthreats.net/open/suricata-7.0.10/emerging.rules.tar.gz.
100% - 5569197/5569197
8/8/2026 -- 05:43:24 - <Info> -- Done.
8/8/2026 -- 05:43:24 - <Info> -- Loading distribution rule file /etc/suricata/rules/app-layer-events.rules
8/8/2026 -- 05:43:24 - <Info> -- Loading distribution rule file /etc/suricata/rules/decoder-events.rules
8/8/2026 -- 05:43:24 - <Info> -- Loading distribution rule file /etc/suricata/rules/dhcp-events.rules
8/8/2026 -- 05:43:24 - <Info> -- Loading distribution rule file /etc/suricata/rules/dnp3-events.rules
8/8/2026 -- 05:43:24 - <Info> -- Loading distribution rule file /etc/suricata/rules/dns-events.rules
8/8/2026 -- 05:43:24 - <Info> -- Loading distribution rule file /etc/suricata/rules/files.rules
8/8/2026 -- 05:43:24 - <Info> -- Loading distribution rule file /etc/suricata/rules/http2-events.rules
8/8/2026 -- 05:43:24 - <Info> -- Loading distribution rule file /etc/suricata/rules/http-events.rules
8/8/2026 -- 05:43:24 - <Info> -- Loading distribution rule file /etc/suricata/rules/ipsec-events.rules
8/8/2026 -- 05:43:24 - <Info> -- Loading distribution rule file /etc/suricata/rules/kerberos-events.rules
8/8/2026 -- 05:43:24 - <Info> -- Loading distribution rule file /etc/suricata/rules/modbus-events.rules
8/8/2026 -- 05:43:24 - <Info> -- Loading distribution rule file /etc/suricata/rules/mqtt-events.rules
8/8/2026 -- 05:43:24 - <Info> -- Loading distribution rule file /etc/suricata/rules/nfs-events.rules
8/8/2026 -- 05:43:24 - <Info> -- Loading distribution rule file /etc/suricata/rules/nntp-events.rules
8/8/2026 -- 05:43:24 - <Info> -- Loading distribution rule file /etc/suricata/rules/quic-events.rules
8/8/2026 -- 05:43:24 - <Info> -- Loading distribution rule file /etc/suricata/rules/rfb-events.rules
8/8/2026 -- 05:43:24 - <Info> -- Loading distribution rule file /etc/suricata/rules/smb-events.rules
8/8/2026 -- 05:43:24 - <Info> -- Loading distribution rule file /etc/suricata/rules/smtp-events.rules
8/8/2026 -- 05:43:24 - <Info> -- Loading distribution rule file /etc/suricata/rules/ssh-events.rules
8/8/2026 -- 05:43:24 - <Info> -- Loading distribution rule file /etc/suricata/rules/stream-events.rules
8/8/2026 -- 05:43:24 - <Info> -- Loading distribution rule file /etc/suricata/rules/tls-events.rules
8/8/2026 -- 05:43:24 - <Info> -- Ignoring file f625293e2432dbf07497d06349de6f0b/rules/emerging-deleted.rules
8/8/2026 -- 05:43:26 - <Info> -- Loaded 68175 rules.
8/8/2026 -- 05:43:27 - <Info> -- Disabled 15 rules.
8/8/2026 -- 05:43:27 - <Info> -- Enabled 0 rules.
8/8/2026 -- 05:43:27 - <Info> -- Modified 0 rules.
8/8/2026 -- 05:43:27 - <Info> -- Dropped 0 rules.
8/8/2026 -- 05:43:27 - <Info> -- Enabled 136 rules for flowbit dependencies.
8/8/2026 -- 05:43:27 - <Info> -- Backing up current rules.
8/8/2026 -- 05:43:31 - <Info> -- Writing rules to /var/lib/suricata/rules/suricata.rules: total: 68175; enabled: 52235; added: 90; removed: 1; modified: 1354
8/8/2026 -- 05:43:32 - <Info> -- Writing /var/lib/suricata/rules/classification.config
8/8/2026 -- 05:43:32 - <Info> -- Testing with suricata -T.
```

Step 4 – Configure Suricata Interface Monitoring

Configuration file:

```
sudo nano /etc/suricata/suricata.yaml
```

Modified section:

```
af-packet:
- interface: enp0s8
```

Reason:

The Host-Only adapter receives Nmap traffic generated from the attacker VM. I also added enp0s3 interface so I could test the working.

Validate configuration:

```
sudo suricata -T -c /etc/suricata/suricata.yaml
```

Restart service:

```
sudo systemctl restart suricata
```

Screenshot Placeholder

```
# Linux high speed capture support
af-packet:
  - interface: enp0s3
  - interface: enp0s8
    # Number of receive threads. "auto" uses the number of cores
    #threads: auto
    # Default clusterid. AF_PACKET will load balance packets based on flow.
    cluster-id: 99

root@vbox:/etc/suricata# suricata -T -c /etc/suricata/suricata.yaml
i: suricata: This is Suricata version 7.0.10 RELEASE running in SYSTEM mode
i: suricata: Configuration provided was successfully loaded. Exiting.
root@vbox:/etc/suricata#
```

Step 5 – Enable Eve JSON Logging

Verified the following configuration:

```
outputs:
  - eve-log:
      enabled: yes
      filename: eve.json
```

Log location:

/var/log/suricata/eve.json

Verify:

```
ls -l /var/log/suricata/eve.json
```

Monitor events:

```
tail -f /var/log/suricata/eve.json
```

Screenshot Placeholder

```
"flows_evicted_needs_work":0,"flows_evicted_pkt_inject":0,"flows_evicted":0,"flows_injected":0,"flows_injected_max":0},
"end":{"state":{"new":6,"established":0,"closed":0,"local_bypassed":0,"capture_bypassed":0},"tcp_state":{"none":0,"syn_s
ent":0,"syn_recv":0,"established":0,"fin_wait1":0,"fin_wait2":0,"time_wait":0,"last_ack":0,"close_wait":0,"closing":0,"c
losed":0},"tcp_liberal":0},"mgr":{"full_hash_pass":74,"rows_per_sec":6553,"rows_maxlen":1,"flows_checked":18,"flows_noti
meout":12,"flows_timeout":6,"flows_evicted":6,"flows_evicted_needs_work":0},"spare":9806,"emerg_mode_entered":0,"emerg_m
ode_over":0,"recycler":{"recycled":6,"queue_avg":0,"queue_max":1},"memuse":7234304,"defrag":{"ipv4":{"fragments":0,"rea
ssembled":0},"ipv6":{"fragments":0,"reassembled":0},"max_frag_hits":0},"flow_bypassed":{"local_pkts":0,"local_bytes":0,"
local_capture_pkts":0,"local_capture_bytes":0,"closed":0,"pkts":0,"bytes":0},"detect":{"engines":[{"id":0,"last_reload":
"2026-08-08T05:46:05.441958+0000"},"rules_loaded":52149,"rules_failed":0,"rules_skipped":0]},"alert":0,"alert_queue_overf
low":0,"alerts_suppressed":1},"app_layer":{"flow":{"http":0,"ftp":0,"smtp":0,"tls":1,"ssh":1,"imap":0,"smb":0,"dcerpc_tc
p":0,"dns_tcp":0,"nfs_tcp":0,"ntp":1,"ftp-data":0,"tftp":0,"ike":0,"krb5_tcp":0,"quic":0,"dhcp":1,"snmp":0,"sip":0,"rfb"
:0,"mqtt":0,"telnet":0,"rdp":0,"http2":0,"bittorrent-dht":0,"failed_tcp":0,"dcerpc_udp":0,"dns_udp":1,"nfs_udp":0,"krb5_
udp":0,"failed_udp":2},"tx":{"http":0,"ftp":0,"smtp":0,"tls":0,"ssh":0,"imap":0,"smb":0,"dcerpc_tcp":0,"dns_tcp":0,"nfs_
tcp":0,"ntp":1,"ftp-data":0,"tftp":0,"ike":0,"krb5_tcp":0,"quic":0,"dhcp":6,"snmp":0,"sip":0,"rfb":0,"mqtt":0,"telnet":0
,"rdp":0,"http2":0,"bittorrent-dht":0,"dcerpc_udp":0,"dns_udp":4,"nfs_udp":0,"krb5_udp":0},"error":{"http":{"gap":0,"all
oc":0,"parser":0,"internal":0},"ftp":{"gap":0,"alloc":0,"parser":0,"internal":0},"smtp":{"gap":0,"alloc":0,"parser":0,"i
nternal":0},"tls":{"gap":0,"alloc":0,"parser":0,"internal":0},"ssh":{"gap":0,"alloc":0,"parser":0,"internal":0},"imap":{"
gap":0,"alloc":0,"parser":0,"internal":0},"smb":{"gap":0,"alloc":0,"parser":0,"internal":0},"dcerpc_tcp":{"gap":0,"allo
c":0,"parser":0,"internal":0},"dns_tcp":{"gap":0,"alloc":0,"parser":0,"internal":0},"nfs_tcp":{"gap":0,"alloc":0,"parser
":0,"internal":0},"ntp":{"gap":0,"alloc":0,"parser":0,"internal":0},"ftp-data":{"gap":0,"alloc":0,"parser":0,"internal":
0},"tftp":{"gap":0,"alloc":0,"parser":0,"internal":0},"ike":{"gap":0,"alloc":0,"parser":0,"internal":0},"krb5_tcp":{"gap
":0,"alloc":0,"parser":0,"internal":0},"quic":{"gap":0,"alloc":0,"parser":0,"internal":0},"dhcp":{"gap":0,"alloc":0,"par
ser":0,"internal":0},"snmp":{"gap":0,"alloc":0,"parser":0,"internal":0},"sip":{"gap":0,"alloc":0,"parser":0,"internal":0
},"rfb":{"gap":0,"alloc":0,"parser":0,"internal":0},"mqtt":{"gap":0,"alloc":0,"parser":0,"internal":0},"telnet":{"gap":0
},"alloc":0,"parser":0,"internal":0},"rdp":{"gap":0,"alloc":0,"parser":0,"internal":0},"http2":{"gap":0,"alloc":0,"parser
":0,"internal":0},"bittorrent-dht":{"gap":0,"alloc":0,"parser":0,"internal":0},"failed_tcp":{"gap":0},"dcerpc_udp":{"all
oc":0,"parser":0,"internal":0},"dns_udp":{"alloc":0,"parser":0,"internal":0},"nfs_udp":{"alloc":0,"parser":0,"internal":
0},"krb5_udp":{"alloc":0,"parser":0,"internal":0},"expectations":0},"memcap_pressure":5,"memcap_pressure_max":5,"http":
0},"memuse":0,"memcap":0},"ftp":{"memuse":0,"memcap":0},"file_store":{"open_files":0}}}
```

Step 6 – Create Custom Nmap Detection Rule

Created a local rule file:

```
sudo nano /etc/suricata/rules/local.rules
```

Added the following rule:

```
alert tcp any any -> $HOME_NET any \
(msg:"LOCAL NMAP SYN SCAN"; \
flags:S; \
threshold:type both, track by_src, count 20, seconds 10; \
classtype:attempted-recon; \
sid:1000001; rev:1;)
```

Rule Explanation:

Parameter	Purpose
flags	Detect SYN packets
count 20	Trigger after 20 packets
seconds 10	Within 10 seconds
classtype	Reconnaissance activity
sid	Unique rule identifier

Validated rule syntax:

Then

Copy the suricata.rules file from /var/lib/suricata/ to /etc/suricata/rules

And

naon /etc/suricata/suricta.yml

Drag down for and long time and find

```
##  
## Configure Suricata to load Suricata-Update managed rules.  
##  
  
default-rule-path: /var/lib/suricata/rules  
  
rule-files:  
  - suricata.rules
```

Change it to :

```
##  
## Configure Suricata to load Suricata-Update managed rules.  
##  
  
default-rule-path: /etc/suricata/rules  
  
rule-files:  
  - suricata.rules  
  - local.rules
```

Then

sudo suricata -T -c /etc/suricata/suricata.yaml

Restarted Suricata:

sudo systemctl restart suricata

Screenshot Placeholder



```
vboxuser@vbox: ~  
GNU nano 8.4 /etc/suricata/rules/local.rules  
alert tcp any any -> $HOME_NET any \  
(msg:"LOCAL NMAP SYN SCAN"; \  
flags:S; \  
threshold:type both, track by_src, count 20, seconds 10; \  
classtype:attempted-recon; \  
sid:1000001; rev:1;)
```

Step 7 – Configure Wazuh Agent

Edited Wazuh configuration:

```
sudo nano /var/ossec/etc/ossec.conf
```

Added:

```
<localfile>  
  <log_format>json</log_format>  
  <location>/var/log/suricata/eve.json</location>  
</localfile>
```

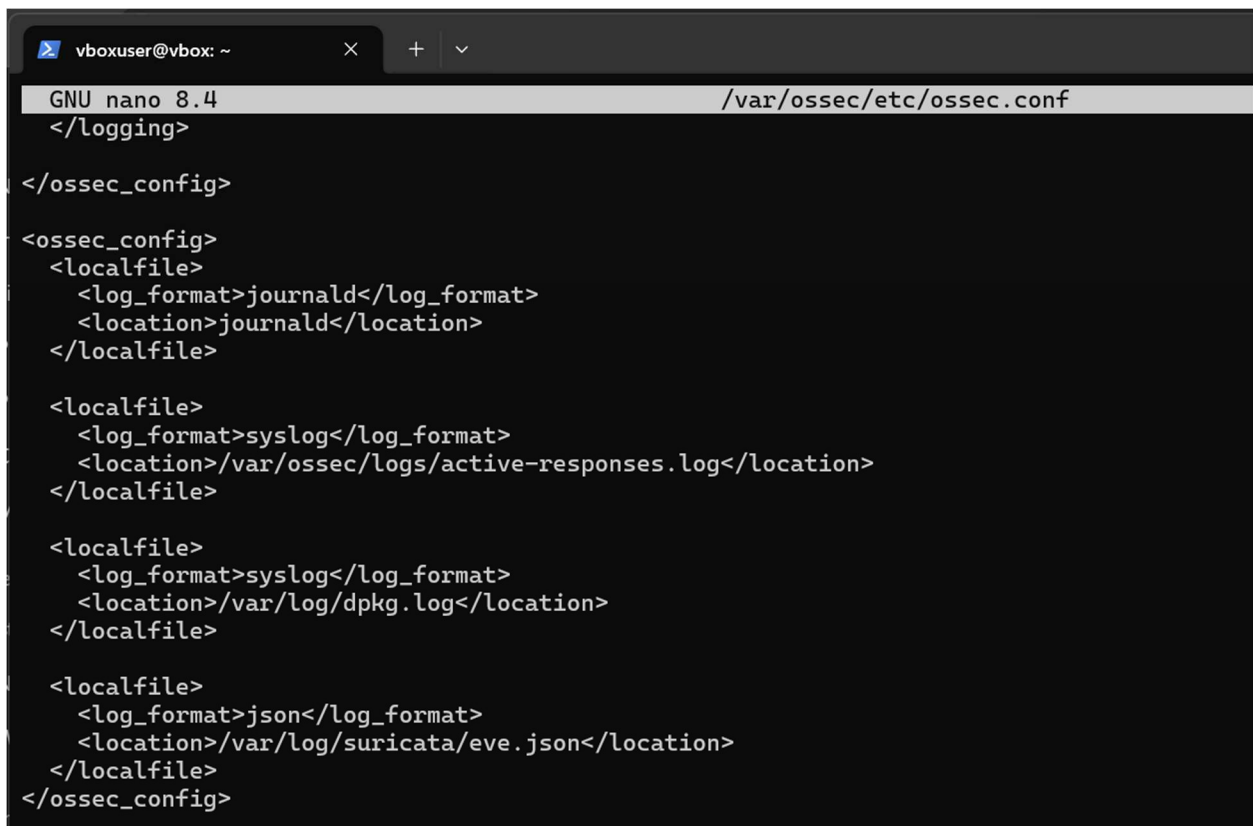
Restarted agent:

```
sudo systemctl restart wazuh-agent
```

Purpose:

- Monitor Suricata JSON logs
- Forward alerts to Wazuh Manager

Screenshot Placeholder



```
vboxuser@vbox: ~  
GNU nano 8.4 /var/ossec/etc/ossec.conf  
</logging>  
  
</ossec_config>  
  
<ossec_config>  
  <localfile>  
    <log_format>journald</log_format>  
    <location>journald</location>  
  </localfile>  
  
  <localfile>  
    <log_format>syslog</log_format>  
    <location>/var/ossec/logs/active-responses.log</location>  
  </localfile>  
  
  <localfile>  
    <log_format>syslog</log_format>  
    <location>/var/log/dpkg.log</location>  
  </localfile>  
  
  <localfile>  
    <log_format>json</log_format>  
    <location>/var/log/suricata/eve.json</location>  
  </localfile>  
</ossec_config>
```

Step 8 – Create Wazuh Detection Rule

On the Wazuh Manager:

```
sudo -i
```

```
sudo nano /var/ossec/etc/rules/local_rules.xml
```

Added:

```
<group name="suricata,nmap,">

  <rule id="100500" level="12">
    <field name="alert.signature">LOCAL NMAP SYN SCAN</field>
    <description>Nmap SYN Scan Detected</description>

    <mitre>
      <id>T1595</id>
    </mitre>

  </rule>

</group>
```

Restarted manager:

```
sudo systemctl restart wazuh-manager
```

Purpose:

- Generate high-priority alerts
- Map activity to MITRE ATT&CK T1595 (Active Scanning)

Screenshot Placeholder

```
GNU nano 8.3 /var/ossec/etc/rules/local_rules.xml
<group name="local,syslog,sshd,">

  <!--
  Dec 10 01:02:02 host sshd[1234]: Failed none for root from 1.1.1.1 port 1066 ssh2
  -->
  <rule id="100001" level="5">
    <if_sid>5716</if_sid>
    <srcip>1.1.1.1</srcip>
    <description>sshd: authentication failed from IP 1.1.1.1.</description>
    <group>authentication_failed,pci_dss_10.2.4,pci_dss_10.2.5,</group>
  </rule>

</group>

<group name="suricata,nmap,">

  <rule id="100500" level="12">
    <field name="alert.signature">LOCAL NMAP SYN SCAN</field>
    <description>Nmap SYN Scan Detected</description>
    <mitre>
      <id>T1595</id>
    </mitre>
  </rule>

</group>
```

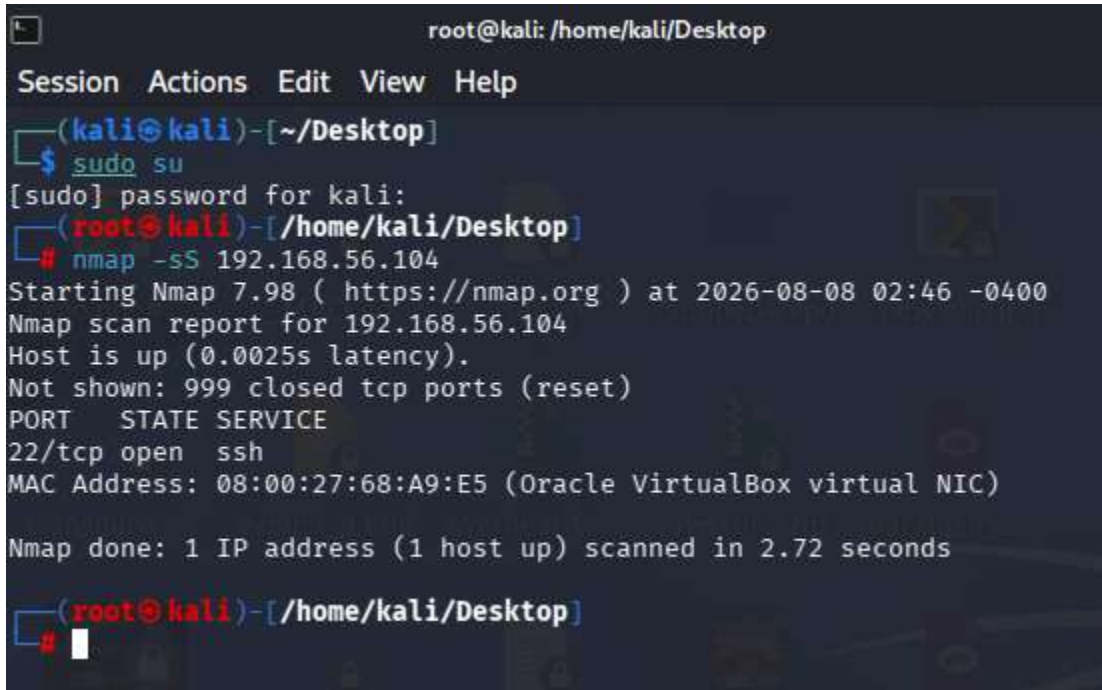
Step 9 – Generate Nmap Scan

From the attacker system:

```
nmap -sS 192.168.56.102
```

This generated SYN packets against the target system.

Screenshot Placeholder

A terminal window titled 'root@kali: /home/kali/Desktop' with a menu bar (Session, Actions, Edit, View, Help). The prompt is '(kali@kali)-[~/Desktop]'. The user enters '\$ sudo su', and the prompt changes to '(root@kali)-[/home/kali/Desktop]'. The user enters '# nmap -sS 192.168.56.104'. The output shows 'Starting Nmap 7.98 (https://nmap.org) at 2026-08-08 02:46 -0400', 'Nmap scan report for 192.168.56.104', 'Host is up (0.0025s latency).', 'Not shown: 999 closed tcp ports (reset)', a table with 'PORT STATE SERVICE' and '22/tcp open ssh', 'MAC Address: 08:00:27:68:A9:E5 (Oracle VirtualBox virtual NIC)', and 'Nmap done: 1 IP address (1 host up) scanned in 2.72 seconds'. The prompt returns to '(root@kali)-[/home/kali/Desktop]'.

```
root@kali: /home/kali/Desktop
Session Actions Edit View Help
(kali@kali)-[~/Desktop]
$ sudo su
[sudo] password for kali:
(root@kali)-[/home/kali/Desktop]
# nmap -sS 192.168.56.104
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-08 02:46 -0400
Nmap scan report for 192.168.56.104
Host is up (0.0025s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 08:00:27:68:A9:E5 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2.72 seconds

(root@kali)-[/home/kali/Desktop]
#
```

Step 10 – Verify Suricata Detection

Search alerts:

```
grep -a "LOCAL NMAP SYN SCAN" /var/log/suricata/eve.json
```

Observed alert:

```
{
  "event_type": "alert",
  "alert": {
    "signature": "LOCAL NMAP SYN SCAN"
  }
}
```

Screenshot Placeholder

```
root@vbox:/etc/suricata# grep -a "LOCAL NMAP SYN SCAN" /var/log/suricata/eve.json
{"timestamp":"2026-08-05T04:38:29.438746+0000","flow_id":1602925739876921,"in_iface":"enp0s8","event_type":"alert","src_ip":"192.168.56.103","src_port":41106,"dest_ip":"192.168.56.104","dest_port":1723,"proto":"TCP","pkt_src":"wire/pcap","alert":{"action":"allowed","gid":1,"signature_id":1000001,"rev":1,"signature":"LOCAL NMAP SYN SCAN","category":"Attempted Information Leak","severity":2},"direction":"to_server","flow":{"pkts_toserver":1,"pkts_toclient":0,"bytes_toserver":60,"bytes_toclient":0,"start":"2026-08-05T04:38:29.438746+0000","src_ip":"192.168.56.103","dest_ip":"192.168.56.104","src_port":41106,"dest_port":1723}}
{"timestamp":"2026-08-05T04:45:00.245744+0000","flow_id":1336939302705999,"in_iface":"enp0s8","event_type":"alert","src_ip":"192.168.56.1","src_port":53199,"dest_ip":"192.168.56.101","dest_port":443,"proto":"TCP","pkt_src":"wire/pcap","alert":{"action":"allowed","gid":1,"signature_id":1000001,"rev":1,"signature":"LOCAL NMAP SYN SCAN","category":"Attempted Information Leak","severity":2},"direction":"to_server","flow":{"pkts_toserver":1,"pkts_toclient":0,"bytes_toserver":66,"bytes_toclient":0,"start":"2026-08-05T04:45:00.245744+0000","src_ip":"192.168.56.1","dest_ip":"192.168.56.101","src_port":53199,"dest_port":443}}
{"timestamp":"2026-08-06T03:06:05.499148+0000","flow_id":1580875782130420,"in_iface":"enp0s8","event_type":"alert","src_ip":"192.168.56.1","src_port":51530,"dest_ip":"192.168.56.101","dest_port":443,"proto":"TCP","pkt_src":"wire/pcap","alert":{"action":"allowed","gid":1,"signature_id":1000001,"rev":1,"signature":"LOCAL NMAP SYN SCAN","category":"Attempted Information Leak","severity":2},"direction":"to_server","flow":{"pkts_toserver":1,"pkts_toclient":0,"bytes_toserver":66,"bytes_toclient":0,"start":"2026-08-06T03:06:05.499148+0000","src_ip":"192.168.56.1","dest_ip":"192.168.56.101","src_port":51530,"dest_port":443}}
{"timestamp":"2026-08-07T04:16:10.610443+0000","flow_id":651508536287455,"in_iface":"enp0s8","event_type":"alert","src_ip":"192.168.56.1","src_port":58746,"dest_ip":"192.168.56.101","dest_port":443,"proto":"TCP","pkt_src":"wire/pcap","alert":{"action":"allowed","gid":1,"signature_id":1000001,"rev":1,"signature":"LOCAL NMAP SYN SCAN","category":"Attempted Information Leak","severity":2},"direction":"to_server","flow":{"pkts_toserver":1,"pkts_toclient":0,"bytes_toserver":66,"bytes_toclient":0,"start":"2026-08-07T04:16:10.610443+0000","src_ip":"192.168.56.1","dest_ip":"192.168.56.101","src_port":58746,"dest_port":443}}
{"timestamp":"2026-08-08T06:46:10.104247+0000","flow_id":729215810266196,"in_iface":"enp0s8","event_type":"alert","src_ip":"192.168.56.103","src_port":44790,"dest_ip":"192.168.56.104","dest_port":587,"proto":"TCP","pkt_src":"wire/pcap","alert":{"action":"allowed","gid":1,"signature_id":1000001,"rev":1,"signature":"LOCAL NMAP SYN SCAN","category":"Attempted Information Leak","severity":2},"direction":"to_server","flow":{"pkts_toserver":1,"pkts_toclient":0,"bytes_toserver":60,"bytes_toclient":0,"start":"2026-08-08T06:46:10.104247+0000","src_ip":"192.168.56.103","dest_ip":"192.168.56.104","src_port":44790,"dest_port":587}}
root@vbox:/etc/suricata#
```

Step 11 – Verify Wazuh Alert

Check alerts:

```
tail -f /var/ossec/logs/alerts/alerts.json
```

Observed:

Rule ID: 100500

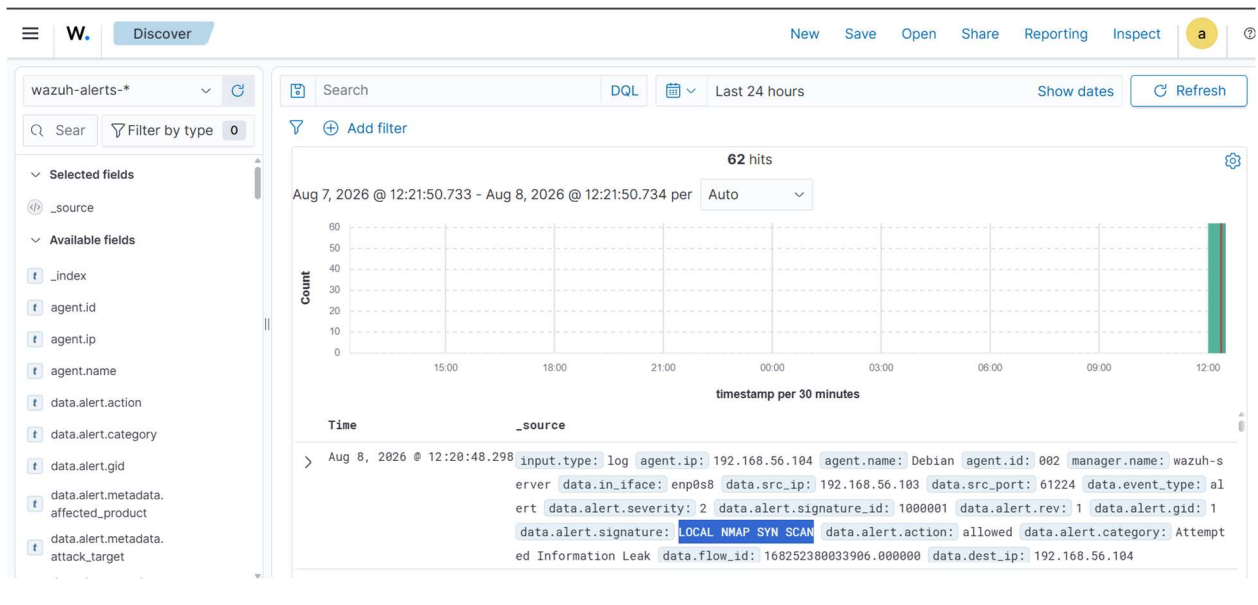
Level: 12

Description: Nmap SYN Scan Detected

Source IP: <Attacker IP>

Dashboard verification confirmed that the alert was received and processed successfully.

Screenshot Placeholder



Troubleshooting Performed

UFW Logging Attempt

Initially, UFW logging was evaluated for Nmap detection.

Issue encountered:

ERROR: problem running `sysctl`

Investigation showed:

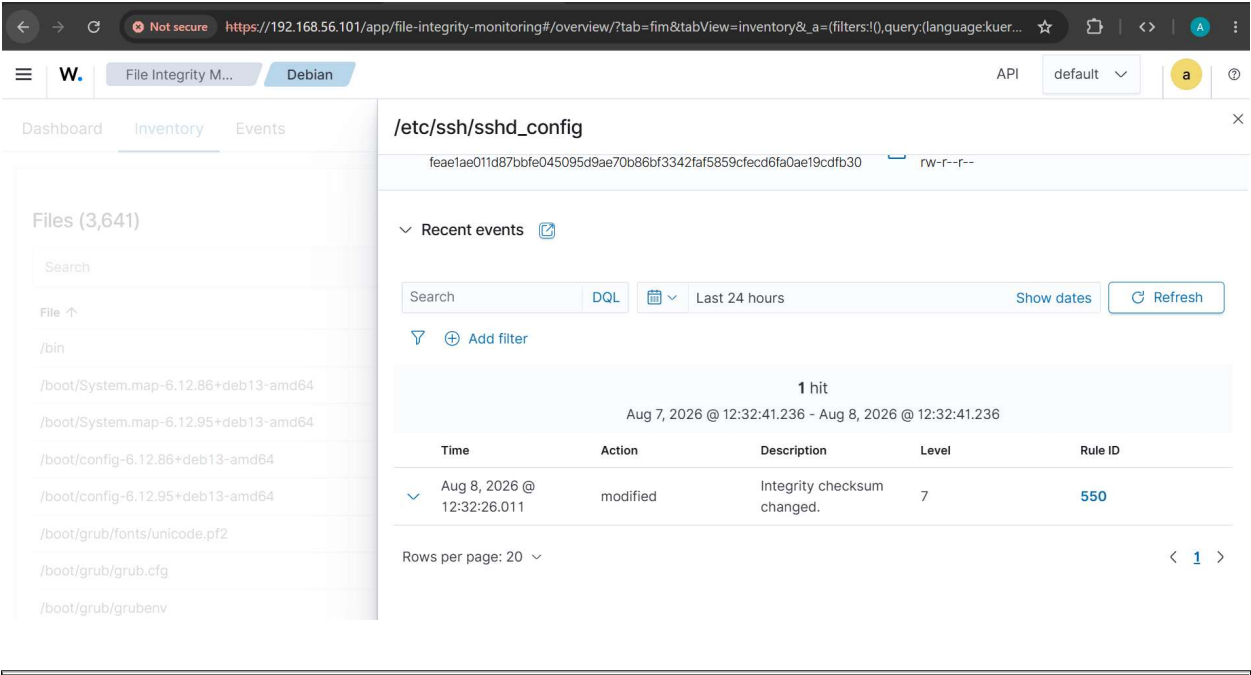
`sudo sysctl --system`

My Debian system didn't have `sysctl` file.

Decision:

- UFW logging was not used.
- Suricata was selected because it provides protocol-aware detection and structured JSON output suitable for Wazuh integration.
- I also performed File integrity monitoring on `/var/ssh/sshd_conf` to perform it I would recommend you to read
- <https://documentation.wazuh.com/current/user-manual/capabilities/file-integrity/advanced-settings.html#who-data-monitoring>

Screenshot Placeholder



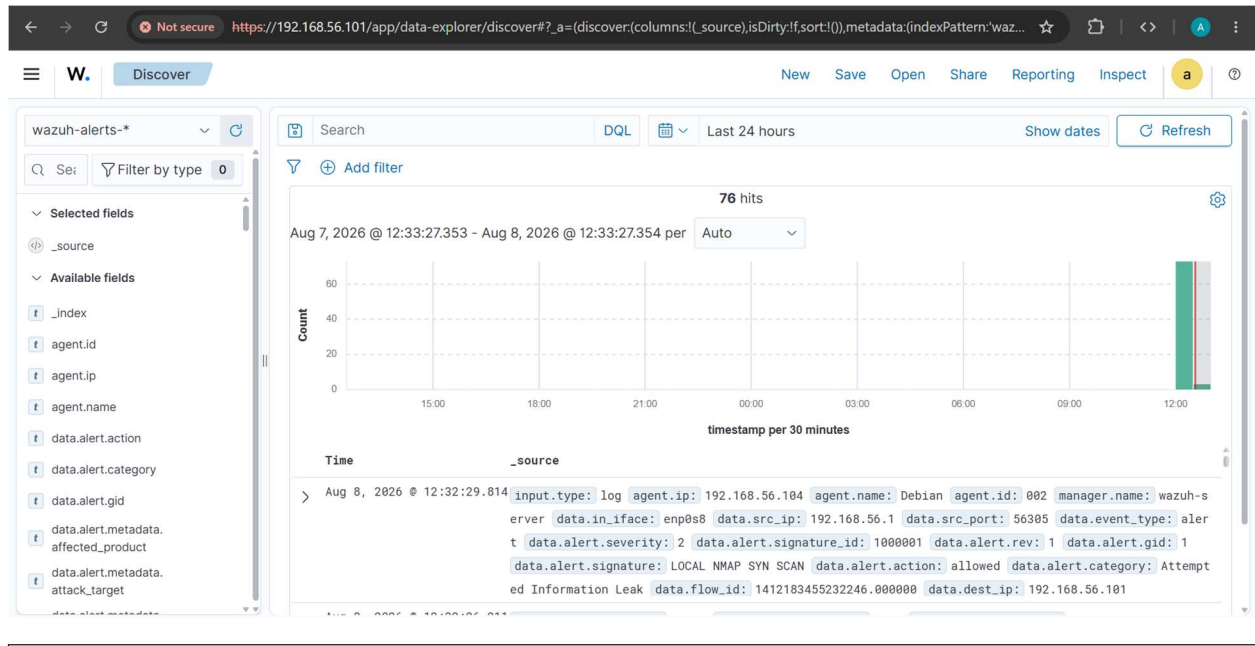
Results

The implementation successfully achieved the objective.

Successful Outcomes

- Suricata installed and operational.
- Emerging Threats rules loaded.
- Custom Nmap SYN detection rule created.
- Suricata generated alerts upon Nmap scanning.
- Wazuh agent collected Suricata events.
- Wazuh manager generated alerts.
- Detection mapped to MITRE ATT&CK framework.
- Alerts visible in the Wazuh Dashboard.

Screenshot Placeholder



Conclusion

A complete Suricata-to-Wazuh integration was successfully implemented on Debian 13. The environment detected Nmap SYN scans in real time and generated corresponding alerts within Wazuh. This approach provides significantly better visibility and detection capabilities than relying solely on firewall logs and forms a strong foundation for network intrusion detection and security monitoring.